

TPC-403: A CRT-Origin Proxy Obstruction for the Signed Deletion Profile

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Abstract

We give an exact adversarial construction for the finite signed coefficient model isolated in TPC-402. For alternating signs on any finite set of primes larger than the window length, the congruences $o \equiv 0 \pmod{p_{2k}}$ and $o \equiv -N \pmod{p_{2k+1}}$ separate the positive and negative mask profiles. Consequently the adjacent raw coefficient is $T_1 P_-$ rather than a quantity controlled by the small global scalar $A_\sigma = P_+ - P_-$. Four exact cases are certified. This is a proxy-level obstruction: no normalized growing bound, arithmetic sign identification, or twin-prime conclusion follows.

1 Finite object

Let $I_o = \{o, o+1, \dots, o+N-1\}$ and $T_{uv} = H^2/(H^2 + (u-v)^2)$. The TPC-402 production identity for an off-diagonal pair is

$$M_\sigma(u, v) = T_{uv}[-A_\sigma + b_\sigma(u) + b_\sigma(v)], \quad A_\sigma = \sum_i \sigma_i a_i, \quad b_\sigma(u) = \sum_{p_i|u} \sigma_i a_i,$$

where $a_i = p_i^3/(Q^2(p_i - 1))$. Here $Q = 8192$, $N = 1024$, and $H = 66$; the signs are the declared synthetic law $\sigma_i = (-1)^i$.

2 CRT construction

Choose distinct primes $p_0 < \dots < p_{2m-1}$ with $p_i > N$. Impose

$$o \equiv 0 \pmod{p_{2k}}, \quad o \equiv -N \pmod{p_{2k+1}}. \quad (1)$$

The Chinese remainder theorem gives a residue class modulo $P = \prod_i p_i$. Adding a sufficiently large multiple of P gives $o > B$ for every prescribed bound B .

For $0 \leq r < N$, an even prime divides $o+r$ only at $r=0$. An odd prime would first divide $o+r$ at $r=N$, outside I_o . Thus

$$b_\sigma(o) = P_+ := \sum_{i \text{ even}} a_i, \quad b_\sigma(o+1) = 0.$$

The pair difference is -1 , so no selected prime divides it. The exact identity therefore gives

$$M_\sigma(o, o+1) = T_1[-(P_+ - P_-) + P_+] = T_1 P_-, \quad P_- := \sum_{i \text{ odd}} a_i. \quad (2)$$

This is an exact raw-coefficient obstruction. In particular, the CRT mask profile is not determined by the global scalar A_σ .

3 Exact certificate

The producer and independent reverse-order replay use the first primes in the fixed shell $8192 < p \leq 16384$ and $B = 10^6$. The results are:

m	$2m$ primes	digits in CRT period	$T_1 P_- / A_\sigma $
1	2	8	411.130774097471742
2	4	16	411.430924042908952
3	6	24	514.632609073478964
4	8	32	374.421682115702654

All CRT residues, masks, and coefficient identities are exact rational equalities. The largest case uses primes 8209, 8219, 8221, 8231, 8233, 8237, 8243, 8263 and has a CRT period with 32 decimal digits. The independent checker uses reverse CRT accumulation and reverse coefficient order; both producer and checker also use a strict response-blind contract stress test.

4 Scope and obstruction

The theorem is parameterized by a finite prime set and an unbounded positive origin class. It does not say that a predeclared bounded interval, including the TPC-400 origins near 7.6×10^6 , contains one of these CRT origins. It also does not lower-bound a locally normalized entry: the geometry denominator can absorb the raw profile. Most importantly, alternating-index signs remain a synthetic modeling choice rather than the arithmetic source sign. Therefore the route ledger is

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The project includes the required README, code, experiments, results, notes, proof package, and PDF. No arithmetic sign theorem or twin-prime result is claimed.